

[DASHBOARD](#) / [I MIEI CORSI](#) / [APPELLI DI CLAUDIO SARTORI](#) / [SEZIONI](#) / [MACHINE LEARNING](#) / [MACHINE LEARNING THEORY](#)

|                        |                                                    |
|------------------------|----------------------------------------------------|
| <b>Iniziato</b>        | Thursday, 13 January 2022, 15:13                   |
| <b>Stato</b>           | Completato                                         |
| <b>Terminato</b>       | Thursday, 13 January 2022, 15:39                   |
| <b>Tempo impiegato</b> | 26 min. 23 secondi                                 |
| <b>Punteggio</b>       | 14,00/15,00                                        |
| <b>Valutazione</b>     | <b>28,00</b> su un massimo di 30,00 ( <b>93%</b> ) |

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the main reason for the *standardization* of numeric attributes?

Scegli un'alternativa:

- ☒ a. Change the distribution of the numeric attributes, in order to obtain gaussian distributions
- ☒ b. Map all the numeric attributes to a new range such that the mean is zero and the variance is one.
- ☐ c. Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence
- ☐ d. Remove non-standard values



Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following *is not* an objective of feature selection

Scegli un'alternativa:

- a. Reduce time and memory complexity of the mining algorithms
- b. Reduce the effect of noise
- ☒ ~~c.~~ c. Select the features with higher range, which have more influence on the computations
- d. Avoid the *curse of dimensionality*

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following types of data allows the use of the euclidean distance?

Scegli un'alternativa:

- a. Ordered data
- b. Document representations
- ☒ ~~c.~~ c. Points in a vector space
- d. Transactional data



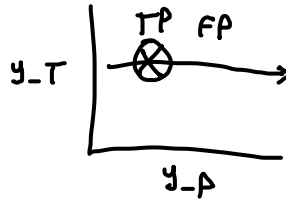
Domanda 4

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Given the definitions below:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives



which of the formulas below computes the *recall* of a binary classifier?

Scegli un'alternativa:

- a.  $TN / (TN + FP)$
- NO** ~~b.~~  $TP / (TP + FP)$
- ~~c.~~  $TP / (TP + FN)$

- d.  $(TP + TN) / (TP + FP + TN + FN)$

Domanda 5

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the *Gini Index*?

Scegli un'alternativa:

- ✓** ~~a.~~ An impurity measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*
- b. An accuracy measure of a dataset alternative to the *Information Gain* and to the *Misclassification Index*
- c. An impurity measure of a dataset alternative to *overfitting* and *underfitting*
- d. A measure of the *entropy* of a dataset



Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

## A Decision Tree is...

**Scegli un'alternativa:**

- a. A tree-structured plan of tests on single attributes to forecast the cluster
- b. A tree-structured plan of tests on multiple attributes to forecast the target
- ☒ ~~a~~ c. A tree-structured plan of tests on single attributes to forecast the target
- d. A tree-structured plan of tests on single attributes to obtain the maximum purity of a node

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

## Which is the main purpose of *smoothing* in Bayesian classification?

**Scegli un'alternativa:**

- a. Dealing with missing values
- ☒ ~~a~~ b. Classifying an object containing attribute values which are missing from some classes in the training set
- c. Classifying an object containing attribute values which are missing from some classes in the test set
- d. Reduce the variability of the data



Domanda 8

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What measure is maximised by the Expectation Maximisation algorithm for clustering?

Scegli un'alternativa:

- ☒ a. The likelihood of an attribute, given the class label
- ☐ b. The likelihood of an example
- ☒ c. The *likelihood* the distributions, defined by the parameters found, given the data available
- ☐ d. The support of a class

Domanda 9

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the statements below is true? (Only one)

Scegli un'alternativa:

- ☐ a. K-means works well also with datasets having a very large number of attributes
- ☐ b. K-means finds the number of clusters which gives the minimum distortion
- ☒ c. Sometimes k-means stops to a configuration which does not give the minimum distortion for the chosen value of the number of clusters.
- ☐ d. K-means always stops to a configuration which gives the minimum distortion for the chosen value of the number of clusters.



Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

After fitting DBSCAN with the default parameter values the results are: 0 clusters, 100% of noise points. Which will be your next trial?

Scegli una o più alternative:

- a. Decrease the radius of the neighborhood
- b. Reduce the minimum number of objects in the neighborhood and the radius of the neighborhood
- ☒ ~~☐~~ Reduce the minimum number of objects in the neighborhood
- ☒ ~~☐~~ Increase the radius of the neighborhood

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following statements regarding the discovery of association rules is true? (One or more)

Scegli una o più alternative:

- ☒ ~~☐~~ The confidence of a rule can be computed starting from the supports of itemsets
- ☒ ~~☐~~ The support of an itemset is anti-monotonic with respect to the composition of the itemset
- c. The support of a rule can be computed given the confidence of the rule
- d. The confidence of an itemset is anti-monotonic with respect to the composition of the itemset



Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

How does *pruning* work when generating frequent itemsets?

Scegli un'alternativa:

- ☐ a. If an itemset is frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated
- ☒ b. If an itemset is not frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated
- ☐ c. If an itemset is frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated
- ☐ d. If an itemset is not frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

Domanda **13**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In *feature selection*, what is the Principal Component Analysis?

- ☐ a. A mathematical technique used to find the principal attributes which determine the classification process
- ☐ b. A heuristic technique used to find a subset of the attributes which produces the same classifier
- ☐ c. A mathematical technique used to transform non numeric attributes into numeric attributes
- ☒ d. A mathematical technique used to transform a set of numeric attributes into a smaller set of numeric attributes which capture most of the variability in data

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

How can we measure the quality of a trained regression model?

- a. Counting the number of values correctly forecast
- b. With a confusion matrix
- ☒ ~~X~~ c. With a formula elaborating the difference between the forecast values and the true ones
- d. With precision, recall and accuracy

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is different from the others?

Scegli un'alternativa:

- ☒ ~~X~~ a. Dbscan
- b. SVM
- c. Decision Tree
- d. Neural Network









Vai a...

[Machine Learning - Python Lab](#) ►

